

From Endoscopic Image Sequences to a Simulable Soft-Tissue Digital Twin: Gaussian Surface Tracking and Force-Scale-Aware Mechanical Inversion

Ranyang Li · Haotian Ma · Zhipeng Lin · Nan Wei

Abstract Endoscopic reconstruction estimates appearance and surface motion but does not by itself provide a mechanically valid, simulable tissue model. We propose a soft-tissue digital-twin framework that converts endoscopic images into a co-registered Gaussian surface, optical material representation, displacement field, and force-aware volumetric mechanics. A canonical anisotropic Gaussian field is reconstructed with depth-valid, instrument-excluded supervision and tracked by fixed primitive identities using graph regularization, constant-velocity initialization, and divergence rollback. An explicit Gaussian-to-tetrahedral interface transfers surface observations to a differentiable Neo-Hookean solver. Known forces permit modulus inversion; without force sensing, a derived force-scale relation limits inference to stiffness contrast. Each parameter is labelled measured, estimated, or assumed and accompanied by uncertainty. Evaluation combines two EndoNeRF sequences, two SCARED keyframes, 60 finite-element scenarios, six phantom acquisitions, and one clinical computed-tomography case. On a strict 28-frame holdout, tissue peak signal-to-noise ratio/structural similarity were 16.7 dB/0.564 for the present method and 36.6 dB/0.939 for EndoGaussian. Across 60 finite-element scenarios, median errors were 9.6% for background modulus, 32.0% for inclusion modulus, and 32.4% for contrast. Phantom contrast was $3.80\times$ versus $3.56\times$ by compression testing. The contribution is a traceable observation-to-simulation interface; absolute tissue modulus still requires measured force.

Keywords digital twin · soft tissue · endoscopic reconstruction · elasticity inversion · parameter provenance · biomechanics

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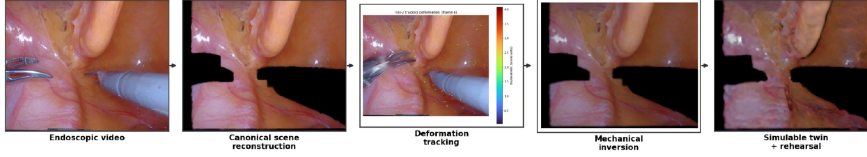
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1 Introduction

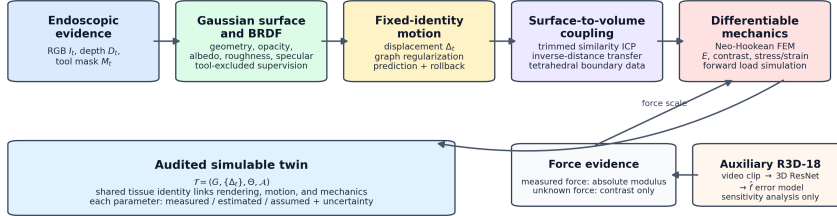
Surgical rehearsal and patient-specific intervention planning require more than a visual reconstruction: the virtual tissue must also respond to a prescribed load through a mechanically interpretable forward model [16, 9]. Endoscopic reconstruction can recover appearance, surface geometry, and motion, but these observations do not by themselves identify Young’s modulus, Poisson’s ratio, or stress. This work therefore addresses the interface between endoscopic observation and a simulable soft-tissue representation.

The central difficulty is evidential. Geometry and reflectance are constrained by images and auxiliary depth, whereas elasticity is constrained only through deformation under a known or otherwise calibrated load. Values copied from a tissue table remain assumptions rather than patient-specific measurements. The proposed twin consequently attaches a provenance state (measured, estimated, or assumed) and uncertainty to each physical parameter, so that simulated stress and strain can be interpreted according to the evidence that supports them.

(a) Observable sequence and twin outputs



(b) Co-registered computational graph and force-dependent identifiability



Core contribution: an explicit observation-to-simulation interface; the renderer, R3D-18 backbone, and constitutive law are established components.

Fig. 1 Overall soft-tissue digital-twin framework. (a) Endoscopic observations are reconstructed, tracked, mechanically parameterized, and driven forward for rehearsal. (b) The computational graph keeps Gaussian surface identity fixed while transferring displacement to a separate tetrahedral discretization. Force evidence determines whether absolute modulus or only stiffness contrast is identifiable. The compact R3D-18 branch summarizes the auxiliary estimator used only to define a force-error model; it is external to the core twin.

1.1 Related work and methodological gap

EndoNeRF [18] and EndoSurf [21] reconstruct deforming endoscopic scenes with neural radiance or signed-distance fields. EndoGaussian [12] uses anisotropic Gaussians and a learned spatiotemporal deformation field to achieve high-quality, real-time novel-view synthesis. Related inverse-rendering methods decompose geometry, lighting, and bidirectional reflectance [11, 5], while deformable SLAM and robust pose estimation address camera motion [8, 6]. These methods establish visual reconstruction but do not produce force-conditioned tissue mechanics.

Physics-aware neural representations address complementary parts of the problem. PAC-NeRF identifies material parameters from multi-view motion under known loading [10]; PhysDreamer estimates relative stiffness without an absolute force scale [22]; and PhysGaussian and Spring-Gaus couple visual primitives to prescribed or learned dynamics [20, 23]. Quasi-static elastography likewise shows that deformation can support stiffness contrast under a shared-load approximation, while absolute modulus requires force information [14, 19, 17]. No compared method provides the complete endoscopic observation-to-volumetric-simulation interface together with explicit parameter provenance.

We propose a co-registered representation $\mathcal{T} = (G, \{\Delta_t\}, \Theta, \mathcal{A})$. A canonical Gaussian field G stores geometry and Cook–Torrance optical material; fixed primitive identities carry displacement Δ_t through time; an explicit alignment and interpolation map transfers surface observations to a tetrahedral Neo-Hookean model with parameters Θ ; and $\mathcal{A}$ records provenance and uncertainty. Constant-velocity initialization, graph regularization, and divergence rollback stabilize tracking. A force-scale analysis establishes that a common scaling of force and regional moduli leaves equilibrium displacement and stiffness contrast unchanged, which prevents unsupported absolute-modulus claims when force is not measured.

The evaluation uses two EndoNeRF sequences, two SCARED keyframes, all 60 scenarios of a four-load finite-element benchmark, six ultrasound-phantom acquisitions with compression-test references, and one de-identified clinical CT geometry. Under the same 28-frame holdout, EndoGaussian reaches 36.6 dB tissue PSNR and 0.939 tissue SSIM, compared with 16.7 dB and 0.564 for the present representation; reconstruction superiority is therefore not claimed. The complete FEM campaign gives median errors of 9.6% for background modulus, 32.0% for inclusion modulus, and 32.4% for contrast. The phantom contrast is $3.80\times$ versus $3.56\times$ by compression testing. These experiments test the integrated twin and its identifiability limits rather than clinical effectiveness.

2 Methods

2.1 Study design, data, and evaluation

This computational methods study separates visual reconstruction, surface tracking, known-force inversion, force-uncertain contrast estimation, and geometry-to-volume coupling. EndoNeRF contributes two RGB sequences (63 and 156 frames) with auxiliary depth and instrument masks; SCARED contributes two stereo keyframes with metric depth [18,1]. The synthetic benchmark contains 60 Neo-Hookean scenarios and four loads with ground-truth regional moduli, measured forces, and noisy surface motion. The physical dataset contains six ultrasound indentation acquisitions and independent compression-test moduli [3]. A single de-identified hospital CT case is used only to construct a patient-specific tetrahedral geometry.

The strict reconstruction split excludes frames satisfying $(t - 1) \bmod 8 = 0$ from both methods. PSNR and SSIM use identical source images and masks; tissue SSIM averages the local 11-by-11 SSIM map only over tissue pixels. Mechanical accuracy is reported as relative modulus error, contrast error, and held-out-load displacement error. FEM scenarios are processed in lexical order with a frozen 20-iteration inversion. Medians, means, interquartile ranges, and per-case values are reported; the phantom median uses a nonparametric bootstrap 95% confidence interval.

2.2 Digital-twin representation

The twin is $\mathcal{T} = (G, \{\Delta_t\}, \Theta, \mathcal{A})$, where G is the canonical Gaussian geometry and optical material, Δ_t is fixed-identity surface displacement, Θ contains the mechanical parameters and forward operator, and $\mathcal{A}$ maps every parameter to its provenance and uncertainty. Inputs are images I_t , depth D_t , and tool masks M_t . A twin is considered simulable when the mechanical operator maps a new load to volumetric displacement and derived stress or strain; it is considered audited only when every parameter has a provenance tag.

2.3 Canonical scene reconstruction

Reconstruction builds G . Each Gaussian g_i stores a position $\mathbf{x}_i \in \mathbb{R}^3$, an anisotropic covariance $\Sigma_i = Q(q_i) \text{diag}(s_i^2) Q(q_i)^\top$ with rotation quaternion q_i and scale s_i , an opacity $o_i \in [0, 1]$, a unit normal $\mathbf{n}_i$, and BRDF parameters $(\rho_d, \rho_s, \alpha, m, a)$. A softmax budget regularizes the relative lobe weights by a partition of unity,

$$\rho_d + \rho_s + a = 1, \quad \rho_d, \rho_s, a \geq 0, \quad (1)$$

which prevents unconstrained growth of these fitted coefficients; it is not by itself a proof of global energy conservation under the discretized lighting model.

Images are formed by front-to-back alpha compositing,

$$\hat{I} = \sum_{i=1}^N c_i \alpha_i \prod_{j<i} (1 - \alpha_j), \quad (2)$$

where c_i is the shaded color of g_i and α_i its projected opacity. The shaded color follows a Cook–Torrance microfacet BRDF [2] split into a diffuse and a specular lobe,

$$c_i = \underbrace{\frac{\rho_d}{\pi} E_d}_{\text{diffuse}} + \rho_s \underbrace{\frac{D(\alpha) G_s F}{4 (\mathbf{n} \cdot \mathbf{l})(\mathbf{n} \cdot \mathbf{v})}}_{\text{specular}}, \quad (3)$$

where E_d is the diffuse irradiance, $D(\alpha)$ the GGX normal distribution, G_s the Smith geometry term, F the Schlick Fresnel term, $\mathbf{l}$ the light direction, and $\mathbf{v}$ the view direction. The dot products in the denominator are clamped away from zero on visible front-facing samples, and the specular term is zero for back-facing samples. The highlight-free diffuse image is obtained by setting the specular lobe to zero.

A key endoscopic adaptation concerns the supervision mask. The EndoN-eRF **masks**/ channel marks *tools*, not tissue, and tools have no valid depth. We therefore define the tissue pixel set as

$$\Omega = \{(u, v) : D(u, v) > \epsilon\} \setminus M, \quad (4)$$

with ϵ a small depth-validity threshold, and supervise only on Ω . Supervising on the tool mask instead of Ω degrades reconstruction by ~ 25 dB (Table 3). The reconstruction objective is

$$\mathcal{L}_1 = \lambda_{ph} \mathcal{L}_{photo} + \lambda_d \mathcal{L}_{depth} + \lambda_\alpha \mathcal{L}_\alpha, \quad (5)$$

with non-negative weights $\lambda_{ph}, \lambda_d, \lambda_\alpha$, a masked $L_1 + 0.5 L_2$ photometric term $\mathcal{L}_{photo}$ on Ω , a depth L_1 term $\mathcal{L}_{depth}$ on Ω , and an alpha binary-cross-entropy term $\mathcal{L}_\alpha$ toward the tissue mask. Lighting is an endoscope-co-located directional light plus a strong ambient spherical-harmonic lobe, so appearance approximates albedo; a learnable exposure closes the residual gap.

2.4 Deformation tracking

Tracking estimates $\{\Delta_t\}$ as a per-Gaussian displacement $\Delta_t \in \mathbb{R}^{N \times 3}$ on the canonical topology, keeping correspondences fixed across time. Fixed correspondence is what distinguishes a twin from a per-frame rebuild: the same primitive is followed through time, so its trajectory is a boundary condition. Naive inertia-only tracking (penalizing $\|\Delta_t - \Delta_{t-1}\|^2$) occasionally diverges on ambiguous frames. We therefore optimize

$$\begin{aligned} \Delta_t^* = \arg \min_{\Delta} \quad & \mathcal{L}_{photo}(\Delta) + \lambda_s \sum_{i=1}^N \|\Delta_i - \bar{\Delta}_{\mathcal{N}(i)}\|^2 \\ & + \lambda_a \|\Delta - 2\Delta_{t-1} + \Delta_{t-2}\|^2, \end{aligned} \quad (6)$$

initialized from the constant-velocity extrapolation $\Delta_t^{(0)} = 2\Delta_{t-1} - \Delta_{t-2}$. The second term is a k -nearest-neighbor Laplacian smoothness over the Gaussian graph; the quantity $\bar{\Delta}_{\mathcal{N}(i)}$ is the mean displacement of the neighbors of primitive i . The third term is an acceleration prior that penalizes changes in velocity rather than velocity itself. A per-frame step cap constrains $N^{-1} \sum_i \|\Delta_{t,i}\|_2$ to at most three times the running median of preceding displacement magnitudes, and a divergence rollback repeats optimization at half the learning rate when the final loss exceeds three times the preceding-frame loss. If the criterion remains unmet, Δ_{t-1} is retained and the frame is flagged. The output is a sequence of replayable boundary conditions.

2.5 Coupling tracked Gaussians to mechanics

The optical and mechanical computations use different discretizations. Gaussian primitives represent the observed surface, whereas the FEM benchmark uses a tetrahedral volume mesh. Coupling is therefore performed through surface displacement rather than by treating Gaussians as finite elements.

For an uninstrumented endoscopic sequence, canonical Gaussian positions are partitioned into four spatial regions C_r by k -means. For region r , the mean displacement magnitude is

$$d_r = \frac{1}{|C_r|} \sum_{i \in C_r} \|\Delta_{t,i}\|_2. \quad (7)$$

Under the common-load approximation used in quasi-static elastography, a relative-stiffness proxy is

$$\kappa_r = \frac{\max_s d_s}{d_r + \epsilon_d}, \quad (8)$$

where $\epsilon_d = 10^{-9}$ in the displacement units prevents division by zero and the most compliant region has $\kappa = 1$. This calculation provides regional contrast only. The absolute modulus remains the tissue prior because the endoscopic sequence does not contain a force measurement.

For known-force benchmark experiments, the observed displacement is defined on surface nodes Γ_{obs} of the tetrahedral mesh. The FEM parameters are optimized so that simulated surface displacement matches these observations. The two settings evaluate complementary outputs: co-registered contrast on tracked Gaussians and absolute-modulus inversion on a force-calibrated volumetric mesh.

For the patient-specific geometry experiment, a binary CT liver mask is block-reduced and each occupied voxel is divided into six tetrahedra with shared nodes. The Gaussian surface is initialized to the volume boundary by principal-axis alignment and refined by trimmed similarity iterative closest point registration. For every covered volume-boundary node, the four nearest transformed Gaussians define inverse-distance weights w_{ij} , and tracked

displacement is transferred as

$$\begin{aligned} u_j^{obs} &= \sum_{i \in \mathcal{N}_4(j)} w_{ij} \Delta_i, \\ w_{ij} &= \frac{(d_{ij} + \varepsilon)^{-1}}{\sum_{k \in \mathcal{N}_4(j)} (d_{kj} + \varepsilon)^{-1}}. \end{aligned} \quad (9)$$

Only boundary nodes whose nearest Gaussian lies within 5% of the volume diagonal are marked as observed. This map preserves separate optical and mechanical discretizations while providing explicit correspondences for boundary conditions. Because the Gaussian and CT surfaces are from different subjects and the CT spacing metadata are unavailable, the experiment validates the computational interface, not anatomical registration.

2.6 Mechanical parameter inversion with known force

For known-force cases, mechanical inversion defines Θ and the forward operator $\mathcal{S}$. Soft tissue is modeled as a quasi-static continuum in equilibrium,

$$R(u; E, \nu, f) = f_{int}(u; E, \nu) - f = 0, \quad (10)$$

where f_{int} is the internal elastic force and f is the applied load. We invert E by gradient descent through a differentiable forward solver,

$$E^* = \arg \min_E \sum_t \|u(E, \nu, f_t) - u_t^{obs}\|_{\Gamma_{obs}}^2 + \gamma \mathcal{R}(E), \quad (11)$$

where $u(E, \nu, f_t)$ is the simulated displacement under load f_t , u_t^{obs} the observed deformation, $\gamma \geq 0$ a regularization weight, and $\mathcal{R}(E)$ a regularizer (log-normal prior plus spatial smoothness). Two forward models are used: a mass-spring lattice (structural and shear springs with stiffness $k = E h$, h the lattice spacing) for synthetic closed-loop studies, and a differentiable Neo-Hookean FEM [7, 13, 4] with strain-energy density

$$\Psi(F) = \frac{\mu}{2} (\text{tr}(F^\top F) - 3) - \mu \log J + \frac{\lambda}{2} (\log J)^2, \quad (12)$$

where F is the deformation gradient, $J = \det F$, and $\mu = E/(2(1 + \nu))$, $\lambda = E\nu/((1 + \nu)(1 - 2\nu))$ are the Lamé parameters. The FEM solve uses Newton iteration with load stepping and warm starts, and gradients are obtained by the implicit-function theorem through the equilibrium. A kPa-scale log-normal tissue database (liver, fat, muscle, skin, tumor;) with near-incompressible ν and constitutive-model tags (Neo-Hookean / Mooney-Rivlin [15] / Ogden [13]) provides priors; the tissue class is selected by heuristic optical features (colour and specular response). This classification selects a prior category and does not directly determine the mechanical values.

2.7 Force-scale propagation

The following result provides the theoretical basis for reporting stiffness contrast as a force-scale-independent estimated quantity.

At fixed Poisson ratio, the Neo-Hookean internal force is linear in Young’s modulus. Scaling all regional moduli and the applied force by the same factor therefore leaves equilibrium displacement unchanged, while regional stiffness contrast is invariant to that scale.

2.8 Parameter provenance

Definition 1 (Parameter provenance) A physical parameter θ carries a provenance tag $\tau(\theta) \in \{\text{measured}, \text{estimated}, \text{assumed}\}$ and a confidence $c(\theta) \in [0, 1]$. A parameter is *measured* when it is acquired directly by a calibrated sensor, *estimated* when it is inferred by reconstruction or inverse simulation, and *assumed* when it is taken from a tissue-class table without case-specific evidence.

Young’s modulus uses a log-normal prior $\log E \sim \mathcal{N}(\mu_0, \sigma_0^2)$, appropriate because tissue stiffness spans orders of magnitude and must stay positive. A Gaussian–Gaussian update combines the prior with deformation evidence in closed form,

$$\begin{aligned}\sigma_{post}^{-2} &= \sigma_0^{-2} + \sigma_{obs}^{-2}, \\ \mu_{post} &= \sigma_{post}^2 \left(\frac{\mu_0}{\sigma_0^2} + \frac{\mu_{obs}}{\sigma_{obs}^2} \right).\end{aligned}\tag{13}$$

where $(\mu_{obs}, \sigma_{obs})$ summarize the deformation evidence. The confidence is the fractional posterior narrowing, $c = 1 - \sigma_{post}/\sigma_0$. For uncertainty propagation, a Laplace approximation $p(\log E \mid \text{data}) \approx \mathcal{N}(\log E^*, H^{-1})$ with H the Hessian of the negative log-posterior at the optimum is sampled by Monte Carlo through the simulator to a 95% prediction interval.

2.9 Twin synchronization and simulation

A digital twin is not a one-shot reconstruction but a continuously updated copy [16]. Synchronization is the tracking loop: as each new frame arrives, the tracker updates $\{\Delta_t\}$ on the fixed canonical topology, and mechanical inversion can reuse the accumulated deformation as additional inversion evidence. Once constructed, the twin is used by driving $\mathcal{S}$ with new loads: given a planned instrument trajectory $\{f_t^{plan}\}$, the simulator produces the predicted deformation $\{u(E^*, \nu, f_t^{plan})\}$, which can be rendered through the canonical field for visual rehearsal or post-processed for quantities of interest (contact force, strain energy density, margin strain). The audit modulates how these predictions are presented: predictions are accompanied by intervals propagated from the corresponding parameter uncertainties.

Remark 1 (Optimization and learned components) The Gaussian field is optimized as a set of per-primitive parameters, the renderer is differentiable, and mechanical inversion optimizes physical parameters through the simulator. The auxiliary R3D-18 force estimator is external to the twin and is used only to define the force-error model in Sec. 2.7.

3 Results

3.1 Reconstruction and optical material

Table 1 Strict one-in-eight EndoNeRF holdout using the same 28 frames, tissue masks, and evaluator.

Method	Full PSNR	Full SSIM	Tissue PSNR	Tissue SSIM
Present method	11.71	0.436	16.71	0.564
EndoGaussian	16.61	0.798	36.59	0.939

EndoGaussian exceeds the present method by 19.87 dB tissue PSNR and 0.375 tissue SSIM (Table 1). The gap is consistent across both scenes. EndoGaussian uses a learned temporal deformation field optimized for novel-view synthesis, whereas the present representation retains approximately 6.5–6.8 thousand primitive identities and interpolates held-out displacement to preserve mechanical correspondence. Optical decomposition produces spatially varying albedo, roughness, and specular maps on EndoNeRF and SCARED (Fig. 2); these are estimated image-based quantities, not mechanical measurements.

3.2 Deformation tracking

Fixed-identity tracking follows the contact-centred motion over the tested sequence (Fig. 3). The inertia-only ablation diverges at frame 5 with worst-frame loss 37.23. Constant-velocity initialization, graph regularization, and rollback reduce the corresponding loss to 1.55 and retain the preceding valid state when re-optimization does not satisfy the acceptance criterion.

3.3 Mechanical inversion and physical validation

All 60 predefined FEM scenarios complete under the frozen protocol (Fig. 4a). Background modulus is moderately recovered, but inclusion modulus and contrast remain weak: their median errors exceed 32%, and contrast error reaches 89.3% in the worst case. The held-out-load experiment uses a representative

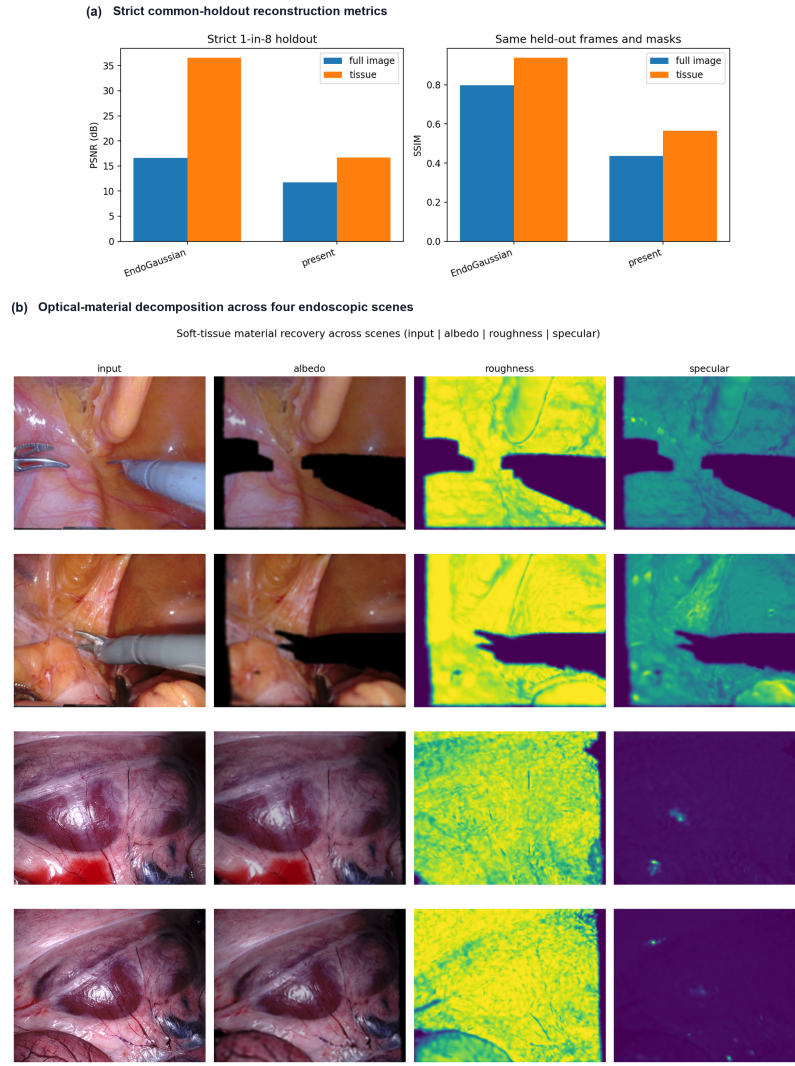


Fig. 2 Reconstruction and optical-material evidence. (a) Strict one-in-eight holdout comparison using identical frames, tissue masks, and metrics. (b) Input, fitted albedo, roughness, and specular components for the two EndoNeRF scenes and two SCARED keyframes. EndoGaussian remains substantially more accurate for novel-view synthesis; the material maps are estimated outputs of the canonical field.

(a) Fixed-identity motion: displacement, rendering, and error

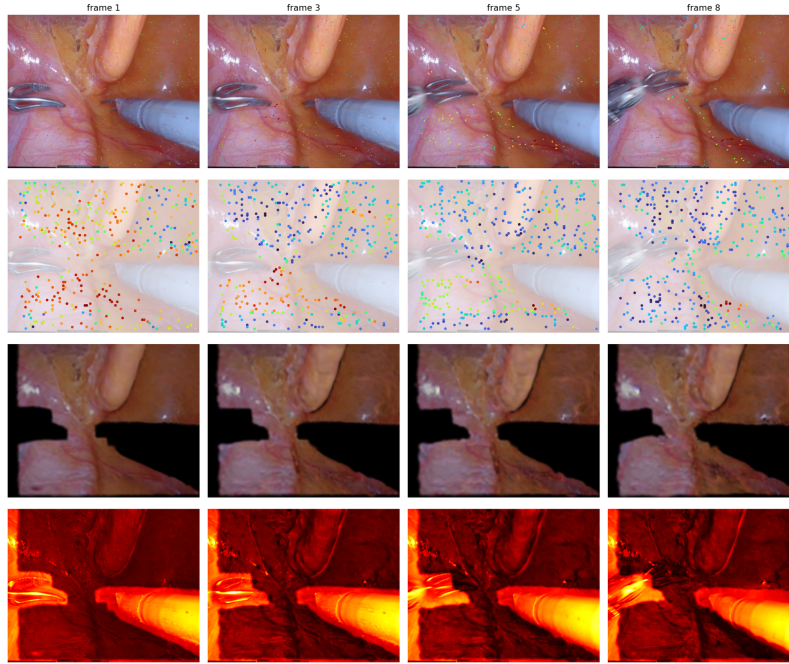


Fig. 3 Fixed-identity deformation tracking on the EndoNeRF pulling sequence. Columns show four times; rows show tracked displacement on the observed frame, displacement magnitude, rendered deformed field, and photometric error. Primitive identity is retained across time to provide co-registered surface observations for mechanics.

Table 2 Quantitative mechanical and interface validation.

Experiment	Evidence	Result
Uniform synthetic loop FEM60 background	known force and E measured force, noisy motion	0.01% modulus error median 9.6%; mean 9.9%
FEM60 inclusion	measured force, sur- face motion	median 32.0%; mean 30.1%
FEM60 contrast Held-out loads	common force scale material fitted on press-left	median 32.4%; mean 33.8% 11.1% / 3.7% / 14.9% displace- ment error
Force-error propagation	auxiliary error model	E_{bg} error $4.1 \pm 4.0\%$; contrast stable
Physical phantom Gaussian-volume map	no absolute force scale cross-subject inter- face	$3.80\times$ vs $3.56\times$; 6.9% error 2,755/7,111 boundary nodes; 38.7% coverage

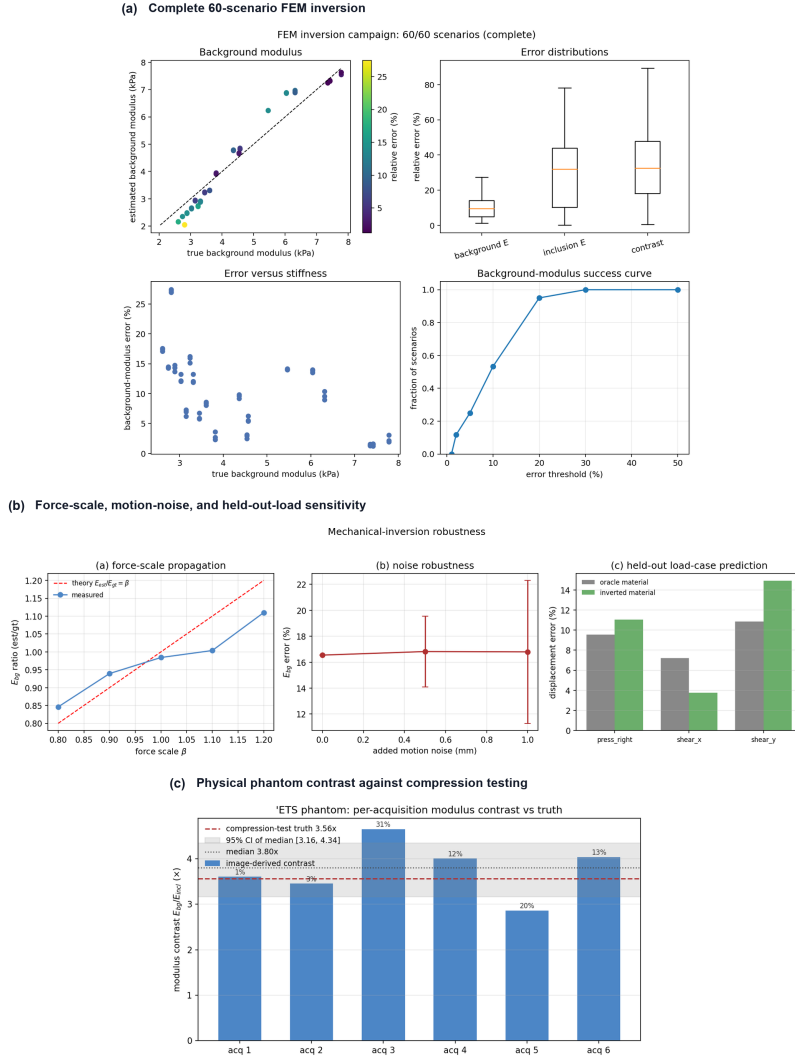


Fig. 4 Mechanical validation. (a) Complete 60-scenario FEM inversion, including modulus agreement, error distributions, stiffness dependence, and the background-modulus success curve. (b) Force-scale, motion-noise, and held-out-load sensitivity. (c) Image-derived phantom stiffness contrast against the independent compression-test reference.

six-scenario subset; material fitted on press-left predicts press-right, shear-x, and shear-y with 11.1%, 3.7%, and 14.9% displacement error, respectively.

The force sweep follows the theoretical scale direction but is not exactly one-to-one because inversion uses regularization, finite load steps, and noisy partial-surface observations (Fig. 4b). The auxiliary R3D-18 error model is used only in this sensitivity test. Without force sensing, the physical phantom

supports contrast rather than absolute modulus: the image-derived median is $3.80\times$ (95% CI 3.16–4.34) against the compression-test reference of $3.56\times$ (Fig. 4c).

3.4 Gaussian surface to patient-specific volume

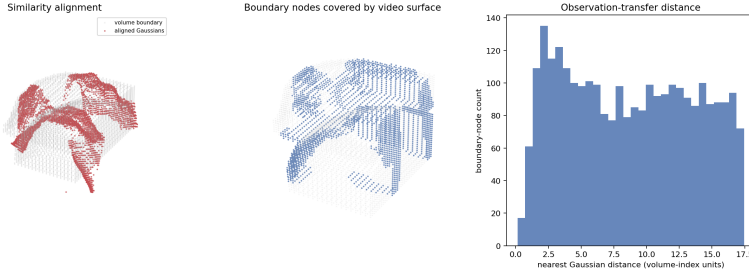


Fig. 5 Cross-subject computational mapping from the tracked Gaussian surface to a CT-derived tetrahedral liver volume. Similarity alignment and inverse-distance interpolation map 2,755 of 7,111 boundary nodes (38.7%). The case tests the software interface, not anatomical correspondence or clinical validity.

The CT mask yields 28,523 nodes and 149,028 tetrahedra. The normalized median surface distance after alignment is 0.0237. Because CT spacing is unavailable and the endoscopic and CT surfaces are from different subjects, neither physical dimensions nor patient-specific material parameters are inferred from this experiment.

3.5 Ablations and uncertainty

Table 3 Controlled component ablations.

Component	Comparison	Result
Tissue supervision	tool / depth-valid mask	11.13 / 36.79 dB
Gaussian count	4,000 / 6,498 / 10,000	35.31 / 36.86 / 38.21 dB
Temporal model	inertia / proposed tracker	37.23 / 1.55 loss
Parameterization	per-node / two-region	$1.69\times$ / $3.19\times$
Forward model	mass-spring / FEM	82% forward-model discrepancy

The Laplace approximation gives $\text{std}(\log E_{bg}) = 1.17$ and $\text{std}(\log E_{incl}) = 21.2$, marking the deep inclusion as weakly identified. Thus, high prediction-interval coverage does not imply that each modulus is accurately estimated; the interval is dominated by the poorly constrained inclusion direction.

4 Discussion

Geometry and reflectance are constrained directly by image and depth observations. In contrast, Young’s modulus is identifiable only when both deformation and loading are available. The force-provenance analysis (Sec. 2.7) confirms that uncertainty in the force scale propagates to absolute modulus but largely cancels in regional stiffness contrast. Absolute modulus and stiffness contrast should therefore be interpreted separately.

Example provenance record For the pulling sequence, albedo and roughness are marked estimated because they are fitted to image observations. Tissue class is also marked estimated because it is selected by an optical heuristic. Young’s modulus is marked assumed (liver prior: median 6 kPa; 95% interval 2.3–16.0 kPa) because no force was instrumented. Regional stiffness contrast is marked estimated from deformation, while Poisson’s ratio and viscosity retain their table-derived assumed labels. Simulations based on this sequence should therefore propagate the prior interval rather than use a single absolute modulus.

The proposed representation could support simulation of alternative instrument loads after reconstruction from an intra-operative video. However, the present evidence establishes technical feasibility rather than clinical utility. Force-instrumented tissue experiments, prospective patient data, and task-specific validation are required before the method can be considered for surgical decision support. Elastography provides the closest clinical analogue because it also estimates stiffness contrast from induced deformation [14]. The phantom experiment indicates agreement with independent compression testing for this relative quantity; it does not establish equivalence with clinical elastography.

Any future clinical implementation would also require institutional data governance, calibrated confidence thresholds, and regulatory evaluation. The provenance map provides a technical mechanism for exposing the evidence associated with each parameter, but does not by itself establish clinical safety.

Reproducibility Source code, evaluation scripts, and data-split definitions are available in the project repository. Every numerical result reported in this paper is produced by the released end-to-end scripts, which cover reconstruction, tracking, and inversion on EndoNeRF and SCARED, the FEM benchmark inversion, the force-provenance analysis, and the phantom audit; each script writes a machine-readable summary and the corresponding figures, from which every number in the paper is drawn.

4.1 Limitations

The strict holdout shows that reconstruction accuracy is substantially below EndoGaussian; fixed surface identity and mechanical coupling do not compensate for the absence of a learned spatiotemporal deformation field. Although

the FEM analysis covers all 60 predefined scenarios, the inclusion modulus and contrast remain weakly recovered. The auxiliary force estimator and its empirical error model are external to the core twin and have not been independently validated beyond their source recordings. The ÉTS phantom uses ultrasound rather than endoscopic colour images, so it validates mechanical contrast but not the optical reconstruction. The mass-spring surrogate has an 82% forward error and is not used to support absolute mechanical estimates on real data. In addition, the clinical CT experiment contains one de-identified case and validates geometry construction only. These limitations preclude claims of clinical performance or absolute-modulus recovery from uninstrumented endoscopic video.

5 Conclusion

The proposed twin couples a canonical Gaussian surface representation with fixed-correspondence tracking, force-aware mechanical parameterization, and parameter provenance. Tracked Gaussians provide co-registered surface motion and regional stiffness contrast for uninstrumented video; absolute modulus is recovered only in known-force FEM experiments. The force-scale identity explains why force uncertainty propagates to absolute modulus but cancels in regional contrast. The common holdout demonstrates that this mechanically coupled representation does not match the reconstruction accuracy of Endo-Gaussian. Across 60 FEM scenarios, median errors are 9.6% for background modulus, 32.0% for inclusion modulus, and 32.4% for contrast; the phantom contrast error is 6.9% against compression testing. The CT-derived tetrahedral mesh and explicit Gaussian-to-volume map demonstrate the software interface on one cross-subject case. These results support the observation-to-simulation framework as a computational method, but do not establish clinical performance, state-of-the-art reconstruction, or force-free absolute-modulus recovery. Future work requires force-instrumented tissue experiments, paired patient-specific surface and volume data, and prospective evaluation of task-specific simulation accuracy.

Abbreviations

BRDF: bidirectional reflectance distribution function; CI: confidence interval; CT: computed tomography; FEM: finite-element method; MPM: material point method; NVS: novel-view synthesis; PSNR: peak signal-to-noise ratio; RGB: red, green, and blue; SCARED: Stereo Correspondence and Reconstruction of Endoscopic Data.

Declarations

Ethics approval and consent to participate

The public EndoNeRF, SCARED, finite-element benchmark, and ÉTS phantom datasets do not involve identifiable human subjects. The hospital computed tomography images are de-identified retrospective scans provided by Henan Provincial People’s Hospital for geometry-source validation; no identifiable patient information is reported. The name of the responsible ethics committee and the approval or waiver reference must be inserted after confirmation by the data-providing institution.

Consent for publication

Not applicable. The manuscript does not contain data from any identified individual person.

Availability of data and materials

Source code, frozen data splits, evaluation scripts, and machine-readable result summaries are available at <https://github.com/liranyang123456-commits/soft-tissue-digital-twin>. EndoNeRF, SCARED, the FEM benchmark, and the ÉTS phantom are public datasets cited in this article. The de-identified hospital CT images are not publicly redistributed under the data-providing institution’s governance.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

RL conceived the study, developed the method, wrote the software, and drafted the manuscript. HM contributed the algorithm, software, and validation. ZL contributed validation, formal analysis, and multi-view evaluation. NW contributed clinical data curation and anatomical review. All authors read and approved the final manuscript.

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